YINFENG ZENG

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EDUCATION

- Ph.D. in Economics, University of Cambridge, the United Kingdom. *Supervisor: Professor Alexei Onatski; Advisor: Professor Oliver Linton. Research areas: econometric theory, random matrix theory, factor models, and financial econometrics. 2022–present*
- MPhil in Finance and Economics, University of Cambridge, the United Kingdom. *Dissertation supervisor: Professor Oliver Linton. Coursework: 83/100 (Lent), 75/100 (Easter). 2021–2022*
- BSc in Statistics, Economics and Finance, University College London, the United Kingdom. *First Class Honours; Dean's List; top 1%. 2018–2021*
- Anglo-Chinese Junior College, Singapore. *SM1 Full Scholarship, Singapore Ministry of Education. 2016–2017*

WORKING PAPERS

1. *Imputation as a Near-Identity Transformation: The Validity of Factor Estimation under Missing Data*. 2026–present.

Abstract. Latent factors are routinely extracted by principal components after a panel's missing entries have been filled — with zeros, means, the EM algorithm, or a purpose-built estimator. Every such choice is a different imputation, and the applied researcher has no way to know whether it has distorted the factors that emerge. We develop a single criterion that resolves the question for all of them at once. For a given panel, the gap between the factor space recovered from the imputed data and that of the unobserved complete data is governed by one quantity: a safety ratio in which the accuracy of the fill and the amount of missingness are weighed against the strength of the weakest factor. Small values preserve the factor space; large ones do not, by a margin we quantify. The criterion is deterministic and free of any assumption on the missingness mechanism, the imputation algorithm, or the pervasiveness of the factors, asking only that the factors be detectable. On this foundation, we prove an exact perturbation law and a central limit theorem for the loss and construct an equivalence certificate that declares an imputation valid only on evidence, with false certification controlled at a chosen level and a feasible estimator that renders it operational. The result is a unified, mechanism-free account of when imputation preserves latent factors.

2. *Common Dynamics in Non-Negative Panels with Many Zeros*, with Ruofan Xu. 2025–present.

Abstract. High-dimensional panels of non-negative time series with frequent zeros — daily rainfall across stations, intraday trading volume across assets, disease counts across regions — combine three features that standard dynamic factor models cannot accommodate together: a point mass at zero, heavy right tails, and strong cross-sectional co-movement. We propose a censored score-driven dynamic factor model that captures all three in a single, likelihood-based framework. Each series is generated by a generalised beta distribution of the second kind (GB2) that is left-shifted and censored at zero, following the univariate construction of Harvey and Ito (2020), while a small number of common latent factors drive the log-scale of every series and are updated recursively by the aggregated cross-sectional score.

3. *Testing Amplifiers in Observed Factor Models*, with Yiyang Zhang. 2025–present.

Abstract. We introduce amplifier factor models, in which the factor loadings of non-amplifier stocks are time-varying functions of the idiosyncratic shocks of designated portfolios called amplifiers. Under a channel-wise exclusion that permits cross-amplification across groups, the amplifier coefficient at a candidate channel (a, k) is estimated by a heteroskedasticity-based instrumental-variables GMM in the spirit of Lewbel (2012), which uses only data already in the model and is consistent under cross-sectional endogeneity between non-amplifier residuals and the amplifying portfolio's innovation. Amplifier candidates are generated either by economic intuition or by a row-masked Group LASSO with selection-consistency and post-selection inference; either route produces a candidate set that is verified by two panel tests — a pervasiveness test and a prevalence test — under

a convergent-normalized-spectrum aggregation theory that accommodates the strong-factor residual structure typical of equity panels. An empirical illustration on U.S. equities is sketched.

4. *Detecting Weak Granulars in Large Panels*, with Zhaocheng Zhang. 2023–present.

Abstract. This paper extends the detection method of Kapetanios et al. (2021) for pervasive units in large panels to allow both the common factors in the factor model and the pervasive units (granulars) to be "weak" — i.e., affecting a considerably large proportion of, rather than the entire, cross section, or with the Gram matrix of the factor loadings not diverging proportionally to the cross-sectional size, N. Empirical illustrations are presented.

5. *Factor Dipping*. 2023–present.

Abstract. Standard principal-component (PC) factor estimation relies on a spectral gap that macroeconomic and financial panels routinely lack. This weak-factor problem, combined with the linear pre-processing transformations implicit in applied work, generates PC uncertainty that standard asymptotic intervals severely understate. We propose factor dipping: re-estimate the factors from bounded, invertible transformations of the data, then invert the transformation. Under the standard specification, the dipped and original estimates span the same space up to $O_p(n^{-2})$ deviations, where n is the diverging spectral gap. We turn this invariance into a specification test with a derived limiting distribution, and into a representation-sensitivity band that quantifies the misspecification-induced uncertainty weak factors inject. Applications to Stock–Watson factors, factor-augmented VAR impulse responses, three-pass risk premia, and quantile factor models show that standard intervals can mislead and downstream structural conclusions can be overturned. We recommend factor dipping as a routine diagnostic before any PC-based inference.

CONFERENCE PRESENTATIONS

- International Symposium on Econometric Theory and Applications (SETA), Tokyo, Japan. Conference presentation; *Factor Dipping*. June 2026.
- Annual Society for Financial Econometrics (SoFiE) Conference, Macau, China. Pre-conference presentation; *Factor Dipping*. June 2026.
- Machine Learning and Financial Econometrics (MLiFE) Conference, Chengdu, China. Conference presentation; *Factor Dipping*. July 2026.
- Asian Summer School of Econometrics and Statistics (ASSES), Beijing, China. Workshop presenter; *Factor Dipping*. July 2026.
- Econometrics PhD Workshop, Faculty of Economics, University of Cambridge. *Common Dynamics in Non-Negative Panels with Many Zeros: A Score-Driven Factor Model*. 2026.
- Econometrics PhD Workshop, Faculty of Economics, University of Cambridge. *Testing Amplifiers in High-Dimensional Observed Factor Models*. 2025.

HONOURS AND AWARDS

- Young Scholar Award, International Symposium on Econometric Theory and Applications (SETA), Tokyo, Japan. 2026.
- Best Paper Award, Asian Summer School of Econometrics and Statistics (ASSES), Beijing, China. 2026.

TEACHING

- F400 Asset Pricing, MPhil in Finance and Economics, University of Cambridge. Teaching Assistant to Professor Christopher Harris; delivered tutorials on risk aversion, CAPM and CCAPM, static and dynamic arbitrage pricing, stochastic calculus, and the Black–Scholes model and exotic option pricing. 2026.
- R300 Advanced Econometrics, MRes/MPhil in Economics, University of Cambridge. Teaching Assistant to Professor Alexei Onatski; delivered tutorials on convergence and asymptotic theory, maximum-likelihood estimation and likelihood-based tests, factor models, panel data, and nonparametric methods. 2023–2024.

- Financial Econometrics Supervision, Centre for Global Programmes, Cambridge Newton College (online). Teaching Assistant; lectured on econometric theory and empirical finance, and supervised group projects and thesis writing. 2023.
- Time-Series Modelling Workshop, Department of Statistical Science, University College London (ChangeMakers programme). Teaching Assistant and Lecturer; prepared teaching materials and lectured undergraduates on time-series modelling and practical model building in R. 2020–2022.